

MA 528: Measure Theory and Integration

Course Code	MA 528
Course Name	Measure Theory and Integration
L-T-P-C	(3-1-0-4)
Prerequisites	MA-511 (Real Analysis)
Intended for	M.Sc./M.S./PhD/B.Tech
Mutual Exclusion	NA
Approval	38th Senate (Feb 2023), in the M.Sc Applied Mathematics curriculum; 50th BoA (Jul 2023)

Course Contents

- **Module 1:** Review of Riemann integral, Algebra of subsets of a non-empty set, Measure on an arbitrary sigma-algebra, Continuity property of measure, The induced outer measure, Measurable sets, Borel Sigma algebra, Monotone class, Completion of a measure space, The Lebesgue measure on $\mathbb{R}$, Properties of Lebesgue measure, Non measurable subsets of $\mathbb{R}$. (14 hours)
- **Module 2:** Simple measurable functions, Integral of non-negative measurable functions, Monotone convergence theorem, Fatou's Lemma, Dominated convergence theorem, Relation between Riemann, Improper and Lebesgue integrals, Riesz-Fischer theorem ($L_1[a,b]$ is a complete metric space), $R[a,b]$ is dense $L_1[a,b]$, Lusin's theorem, L_p - spaces, Convergence of measurable functions (almost everywhere, in measure, in mean). (16 Hours)
- **Module 3:** Absolutely continuous functions, Differentiability of monotone functions (Only statement of Lebesgue-Young theorem), Fundamental theorem of calculus for Lebesgue integrable functions, Radon-Nikodym theorem, Product measure, Fubini's theorem, Signed measure, Riesz representation theorem (Without proof). (12 Hours)

Text Books

1. I. K. Rana, **An introduction to Measure and Integration**, 2nd Edition, Narosa, 2005.
2. G. de Barra, **Measure and Integration**, Wiley Eastern, 1981.

References

1. W. Rudin, **Real and Complex Analysis**, 3rd edition, McGraw-Hill, International Editions, 1987.
2. H. L. Royden, **Real Analysis**, 3rd edition, Prentice-Hall of India, 1985.
3. G. B. Folland, **Real Analysis**, Wiley-Interscience Publication, John Wiley & Sons, 1999.
4. M. Thamban Nair, **Measure and Integration, A first course**, CRC Press, 2020.